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Error in $\begin{cases} \text{outcome} \\ \text{Exposure} \end{cases}$

★ Measurement Error (Attenuation Bias)

- In linear regression, it is assumed that the obs are correctly measured without any error.
- Some variables may not be msbl, then we use proxy variable to measure the effect.

Ex.

$$Y = \beta_0 + \beta_1 A^* + \epsilon^* \quad (\text{ideal})$$

$$A = A^* + \tau, \quad \tau \sim (0, \sigma_\tau^2)$$

↓
random disturbance

$$\begin{aligned} \text{Rewrite } Y &= \beta_0 + \beta_1 (A - \tau) + \epsilon^* \\ &= \beta_0 + \beta_1 A + (\epsilon^* - \beta_1 \tau) \\ &= \beta_0 + \beta_1 A + \epsilon \end{aligned}$$

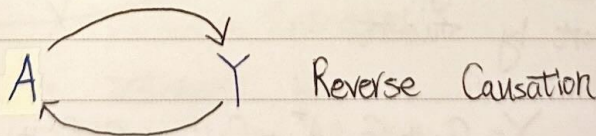
$$\begin{aligned} \text{Hence } E(A\epsilon) &= E[(A^* + \tau)(\epsilon^* - \beta_1 \tau)] \\ &= -\beta_1 E(\tau^2) = -\beta_1 \sigma_\tau^2 \neq 0 \\ \check{\beta}_1 &= \frac{\text{Cov}(Y, A)}{V(A)} = \frac{\text{Cov}(\beta_0 + \beta_1 A^* + \epsilon^*, A^* + \tau)}{V(A)} \end{aligned}$$

$$= \beta_1 \left[\frac{V(A^*)}{V(A^*) + V(\tau)} \right] < \beta_1$$

< 1

★ Simultaneous Bias

- Simultaneity arise when one or more of the predictors is determined by response variable.
- In simple terms



A cause Y and Y cause A.

Ex.

$$\begin{aligned} Y &= \alpha X + u \\ X &= \beta Y + v \end{aligned} \quad \text{by solving equations} \Rightarrow \quad \text{Reduced form} \quad \begin{aligned} Y &= \frac{\alpha v + u}{1 - \alpha \beta} \\ X &= \frac{\alpha u + v}{1 - \alpha \beta} \end{aligned}$$

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$$\text{Cov}(X, u) = \text{Cov}\left(\frac{\alpha u + v}{1 - \alpha\beta}, u\right) = \frac{\alpha}{1 - \alpha\beta} V(u) \neq 0$$

It suffer endogeneity issue.

Ex. (Omitted Variable)

Y : family income

$$\ln(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

X_1 : Husband Education

$$\ln(Y) = \beta_0 + \beta_1 X_1 + \tau \quad (\text{omit } X_2)$$

X_2 : Wife "

	Model 1	Model 2
Const	10.264	10.539
X_1	0.0439	0.0613
X_2	0.0390	

The effect of X_2 has been incorrectly attributed to the X_1 - leading to an overstatement of the latter's importance.

Ex. (Measurement Error)

Y : College GPA

C_1 : high school GPA

C_2 : SAT

$$A = A^* + \tau$$

$\nwarrow$ actual family annual income
 $\nearrow$ report by student.

$$Y = \beta_0 + \beta_A A^* + \beta_1 C_1 + \beta_2 C_2 + \varepsilon^*$$

Using A in place of A^* will bias the OLS estimator of β_A toward zero.

(Remark.

The effect of biasing the coef toward to zero is called attenuation.

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Ex. (Simultaneous Bias)

 Y : murders per capita X : No. of police officers per capita Z : income per capita

- Cities often want to determine how much additional law enforcement will decrease their murders rates.

$$Y = \alpha_1 X + \alpha_2 Z + u$$

- Can we think of police force size as being exogenously determined?

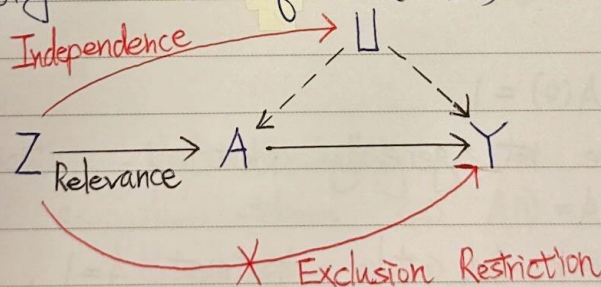
Probably not.

- A city's spending on law enforcement is at least partly determined by its expected murder rates.

$$X = \beta_1 Y + v$$

★ What is remedy of Endogeneity?

⇒ Introducing one or more instrument variables via Two-Stage Least Square (2SLS)

Def. A variable Z is instrument because it meet

1. Relevance, $\text{Cov}(Z, A) \neq 0$ or $Z \not\perp A$
(Z is associated w/ A)

2. Exclusion Restriction, $Y(a, \bar{z}) = Y(a, \bar{z}^*) = Y(a) \quad \forall \bar{z} \text{ and } \bar{z}^* \quad \forall a$
(Z does not affect Y except through its potential effect on A)

3. Independence, $Z \perp U$

(Z is indep of U that affect both A and Y)

The assumption 1 ~ 3 are insufficient for identification

★ 4th IV assumption

a. Effect Homogeneity, $Y(1) - Y(0) \perp\!\!\!\perp Z$

(Constant effect of A on Y across value of covariates)

b. Monotonicity, $A(1) \geq A(0)$ for all individuals (No Defiers)
(Z shift A in same direction for all individuals)

$Z=1$ $Z=0$

★ If we know the value of $A(1)$ and $A(0)$ for each individuals we could classify all individuals into 4 disjoint subpopulations

1. Always-takes,

$$A(1) = A(0) = 1$$

principal strata

2. Never-takes

$$A(1) = A(0) = 0$$

3. Compliers

$$A(1) = 1, A(0) = 0$$

4. Defiers

$$A(1) = 0, A(0) = 1$$

these subpopulation are not generally identified.

• Scenario 1.

Assign to $Z=1$ and take treatment $A=1$, $A(1)=1$
Complier or Always-taker?

• Scenario 2.

Assign to $Z=1$ and take treatment $A=0$, $A(1)=0$
Defier or Never-taker?

★ Identification by Effect Homogeneity
⇕
Constant Effect

$$Y(1) - Y(0) = \beta_0 \Rightarrow Y(1) = Y(0) + \beta_0$$

$$\begin{aligned} \text{Observed } Y &= \begin{cases} Y(1) & \text{if } A=1 \\ Y(0) & \text{if } A=0 \end{cases} \\ &= Y(0) + \{Y(1) - Y(0)\}A \\ &= Y(0) + \beta_0 A \end{aligned}$$

When Z is instrument, under Exclusion Restriction
We have $Y(0) \perp\!\!\!\perp Z$

$$\Rightarrow E(Y(0) | Z=1) = E(Y(0) | Z=0)$$

It implies

$$E(Y - \beta_0 A | Z=1) = E(Y - \beta_0 A | Z=0)$$

$$\Rightarrow E(Y | Z=1) - \beta_0 E(A | Z=1)$$

$$= E(Y | Z=0) - \beta_0 E(A | Z=0)$$

$$\Rightarrow \beta_0 \{E(A | Z=1) - E(A | Z=0)\} = E(Y | Z=1) - E(Y | Z=0)$$

$$\Rightarrow \beta_0 = \frac{E(Y | Z=1) - E(Y | Z=0)}{E(A | Z=1) - E(A | Z=0)}$$

which is the usual IV estimand for dichotomous Instrument.

★ Identification by Monotonicity

Let A as Always-takers $\Leftrightarrow A(1) = A(0) = 1$

N Never-takers $\Leftrightarrow A(1) = A(0) = 0$

C Compliers $\Leftrightarrow A(1) = 1, A(0) = 0$

D Defiers $\Leftrightarrow A(1) = 0, A(0) = 1$

We can decompose causal effect on Y into 4 terms

$$\begin{aligned} E[Y(Z=1) - Y(Z=0)] &= E[Y(Z=1) - Y(Z=0) | A] P(A) \\ &\quad + E[\text{''} | N] P(N) \\ &\quad + E[\text{''} | C] P(C) \\ &\quad + E[\text{''} | D] P(D) \end{aligned}$$

It is weighted average of 4 latent subgroup effects.

i) Given the Exclusion Restriction

$$\begin{aligned} E[Y_{(Z=1)} - Y_{(Z=0)}] &= E[Y_{(Z=1, A(1))} - Y_{(Z=0, A(0))}] \\ &= E[Y_{(A(1))} - Y_{(A(0))}] \end{aligned}$$

Thus,

$$E[Y_{(Z=1)} - Y_{(Z=0)} | A] = 0$$

$$E[Y_{(Z=1)} - Y_{(Z=0)} | N] = 0$$

ii) Given Monotonicity to rule out Defiers

$$E[Y_{(Z=1)} - Y_{(Z=0)}] = E[Y_{(1)} - Y_{(0)} | C] P(C)$$

$\uparrow$
 $A(1)=1, A(0)=0$

iii) Identification of $P(C)$

Consider causal effect Z on A

$$\begin{aligned} E[A_{(1)} - A_{(0)}] &= E[A_{(1)} - A_{(0)} | A] P(A) \\ &\quad + E[\quad \quad | N] P(N) \\ &\quad + E[\quad \quad | C] P(C) \\ &\quad + E[\quad \quad | D] P(D) \\ &= P(C) \end{aligned}$$

Under independence assumption

$$E[A_{(1)} - A_{(0)}] = P(A_{(1)} - A_{(0)} = 1)$$

$$\triangleq P(A=1 | Z=1) - P(A=1 | Z=0)$$

$$E[Y_{(Z=1)} - Y_{(Z=0)}] \triangleq E(Y | Z=1) - E(Y | Z=0)$$

iv) by i) ~ iii)

$$E[Y_{(1)} - Y_{(0)} | C] = \frac{E(Y | Z=1) - E(Y | Z=0)}{P(A=1 | Z=1) - P(A=1 | Z=0)}$$

We identify the effect as the ratio of the diff in means of outcome over diff in means of treatment received.

(Remark. usual IV estimand equals the ave causal effect)
in complier under monotonicity.
 $E[Y(1) - Y(0) | C]$

↑ Village Level

Ex. (Somer-Zeger Vitamin) (CRT) 集群隨機試驗

- Community-based, cluster-randomized trial in Indonesia.
- Evaluate the impact of high-dose vitamin A supplementation on infant survival. → Outcome
- No infant in control village received supplement.

Data N = 23682

Z	A	Y	No. units
0	0	0	74
0	0	1	11514
1	0	0	34
1	0	1	2385
1	1	0	12
1	1	1	9663

$Y=1$ if the infant survived at follow-up period

$A=1$ " received the supplement.

$Z=1$ " village was randomized to receive treatment.

• Analysis

$$P(Y=1 | Z=1) = \frac{2385 + 9663}{34 + 2385 + 12 + 9663} = \frac{12048}{12094}$$

$$P(Y=1 | Z=0) = \frac{11514}{74 + 11514} = \frac{11514}{11588}$$

$$P(A=1 | Z=1) = \frac{12 + 9663}{12094} = \frac{9675}{12094}$$

$$P(A=1 | Z=0) = \frac{0}{11588} = 0$$

then

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$$\frac{P(Y=1|Z=1) - P(Y=1|Z=0)}{P(A=1|Z=1) - P(A=1|Z=0)} = \frac{\frac{12048}{12094} - \frac{11514}{11588}}{\frac{9675}{12094} - 0} = \frac{0.258\%}{0.80\%} \approx 0.323\%$$

the remaining 20% are non-compliers

$\Rightarrow$ Reduction in infant death of 0.323% $\leftarrow$ every 1000 complier infants about +3.23 infants are expected survive due to intervention.

$\frac{2}{2}$ IV in LM.

- In practice, a valid IV may consist of multiple variables or be conti or multi-level.

Consider $Y = \beta_0 + \beta_a A + \beta_u U + \varepsilon$

$$Y = \beta_0 + \beta_a A + \varepsilon^* \quad (\text{Fitted})$$

$$\varepsilon^* = \beta_u U + \varepsilon$$

$$\text{Cov}(Y, Z) = \beta_a \text{Cov}(A, Z) + \text{Cov}(\varepsilon^*, Z)$$

then

$$\frac{\text{Cov}(Y, Z)}{\text{Cov}(A, Z)} = \beta_a + \frac{\text{Cov}(\varepsilon^*, Z)}{\text{Cov}(A, Z)} \quad (Z \perp \varepsilon^* \text{ by Exclusion Restriction})$$

$$= \beta_a$$

(Recall that in Simple linear Reg)

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$$\beta_1 = \frac{\text{Cov}(Y, X)}{V(X)}$$

$$Y = \beta_0 + \beta_a [\pi_0 + \pi_z Z + v] + \varepsilon^*$$

$$= \theta_0 + \theta_z Z + u \quad (\text{reduced})$$

$$\beta_a = \frac{\frac{\text{Cov}(Y, Z)}{V(Z)}}{\frac{\text{Cov}(A, Z)}{V(Z)}} = \frac{\theta_z}{\pi_z}$$

$$A = \pi_0 + \pi_z Z + v$$

★ Two Stage Least Square (2SLS)

- A computational method for calculating the causal effect in two-stage proc.

- The underlying true model are assumed to be linear

$$Y = \beta_0 + \beta_1 A + \beta_2 U + \varepsilon$$

$$A = \pi_0 + \pi_1 Z + \pi_2 U + v$$

★ Two-Stage Proc.

Stage 1: $A \sim Z$

$$A = \pi_0^A + \pi_1^A Z + v_1 \Rightarrow \text{get } \hat{A}$$

Stage 2: $Y \sim \hat{A}$

$$Y = \pi_0^Y + \pi_1^Y \hat{A} + v_2 \Rightarrow \text{get } \hat{\pi}_1^Y$$

★ Why 2SLS work?

$$E(Y|Z) = E(\beta_0 + \beta_1 A + \beta_2 U + \varepsilon | Z)$$

$$= \beta_0 + \beta_1 E(A|Z) + \beta_2 E(U|Z) + E(\varepsilon|Z)$$

$$\beta_0 + \beta_1 E(U)$$

$$= \beta_0^* + \beta_1 E(A|Z)$$

$$E(U)$$

$$E(\varepsilon) = 0$$

★ General Model

$A_1, \dots, A_k$ Endogenous regressors

$X_1, \dots, X_r$ Included exogenous variables $\rightarrow$ uncorrelated w/ ε

$Z_1, \dots, Z_m$ Instrumental variables $\rightarrow$ Excluded exogenous variables

$\rightarrow$ potentially correlated w/ ε

Def. A parameter is said to be identified if different values of parameter will produce different distⁿ of the data.

In IV regression, the coeffs are identified depends on the relation between

- No. of instruments (m)
- No. of endogenous regressor (k)

$$Y = \beta_0 + \beta_1 A_1 + \dots + \beta_k A_k + \theta_1 X_1 + \dots + \theta_r X_r + \varepsilon$$

The coef $\beta_1, \dots, \beta_k$ are said to be

i) exact identified if $m = k$ (just enough to estimate)

ii) overidentified $\quad \quad \quad >$

iii) underidentified $\quad \quad \quad <$ (Too few to estimate)

1. The k 1st stage model, $j=1, \dots, k$

$$A_j = \alpha_0^j + \alpha_1^j X_1 + \dots + \alpha_r^j X_r + \gamma_1^j Z_1 + \dots + \gamma_m^j Z_m + v^j$$

obtain $\hat{A}_1, \dots, \hat{A}_k$

2. In 2nd Stage

$$Y = \beta_0 + \beta_1 \hat{A}_1 + \dots + \beta_k \hat{A}_k + \theta_1 X_1 + \dots + \theta_r X_r + \varepsilon^*$$

★ Projection Viewpoint.

Let $y = n \times 1$ Outcome vector

$A = n \times (k+1)$ Endogenous regressors

$X = n \times r$ Exogenous regressors

$$y = A\beta + X\theta + \varepsilon$$

$$= D\beta + \varepsilon$$

$Z = n \times m$ Instrumental variables

$\overline{W} = [X \ Z] = n \times (r+m)$ full set of instruments

i) $A \sim \overline{W} \Rightarrow$ get $\hat{A}$

$$\hat{A} = \overline{W}(\overline{W}^T \overline{W})^{-1} \overline{W}^T A \quad \left(\begin{array}{l} \text{Project } A \text{ onto space} \\ \text{spanned by } \overline{W} \end{array} \right)$$

$$= PA$$

ii) $y \sim \hat{A}$ and X

$$\hat{\beta} = ([A \ X]^T P [A \ X])^{-1} [A \ X]^T P y$$

$$= (D^T P D)^{-1} D^T P \varepsilon$$

$$= \beta + (D^T P D)^{-1} D^T P \varepsilon$$

→ unbiased if IVs are valid

Let $H = PD$

$$\hat{\beta} = (H^T D)^{-1} H^T y$$

$$= \beta + (H^T D)^{-1} H^T \varepsilon$$

thm (Frisch-Waugh-Lovell)

Given the specification

$$\underline{y} = \underline{X}\underline{\beta} + \underline{\varepsilon} = [\underline{X}_1 | \underline{X}_2] \begin{bmatrix} \underline{\beta}_1 \\ \underline{\beta}_2 \end{bmatrix} + \underline{\varepsilon}$$

$$= \underline{X}_1 \underline{\beta}_1 + \underline{X}_2 \underline{\beta}_2 + \underline{\varepsilon}$$

where $\underline{X}_1$ is of full rank p_1

$$\text{then } \underline{\hat{\beta}}_2 = (\underline{X}_2 (\underline{I}_n - \underline{P}_1) \underline{X}_2')^{-1} \underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{y}$$

<pf>

$$\underline{y} = \underline{y} - \underline{\hat{y}} + \underline{\hat{y}}$$

$$= \underline{X}_1 \underline{\hat{\beta}}_1 + \underline{X}_2 \underline{\hat{\beta}}_2 + \underline{\hat{\varepsilon}}$$

$$\underline{\hat{\varepsilon}} = (\underline{I}_n - \underline{P}) \underline{y} \quad \text{and} \quad \underline{P} = \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}'$$

multiply both side by $\underline{X}_2' (\underline{I}_n - \underline{P}_1)$ w/ $\underline{P}_1 = \underline{X}_1 (\underline{X}_1' \underline{X}_1)^{-1} \underline{X}_1'$

$$\underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{y} = \underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{X}_1 \underline{\hat{\beta}}_1 + \underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{X}_2 \underline{\hat{\beta}}_2 + \underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{\hat{\varepsilon}}$$

Some facts.

- 1) $(\underline{I}_n - \underline{P}_1) \underline{X}_1 = 0$
- 2) $(\underline{I}_n - \underline{P}_1) (\underline{I}_n - \underline{P}) = \underline{I}_n - \underline{P}$
- 3) $(\underline{I}_n - \underline{P}) \underline{X}_2 = 0$

$$\left(\begin{array}{l} \text{span}\{\underline{X}_2\} \subseteq \text{span}\{\underline{X}\} \\ \text{span}\{\underline{X}_1\} \subseteq \text{span}\{\underline{X}\} \\ \Rightarrow \text{span}\{\underline{X}\}^\perp \subseteq \text{span}\{\underline{X}_1\}^\perp \end{array} \right)$$

It follows that

$$\underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{y} = \underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{X}_2 \underline{\hat{\beta}}_2$$

for which we obtain the expression

$$\underline{\hat{\beta}}_2 = (\underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{X}_2)^{-1} \underline{X}_2' (\underline{I}_n - \underline{P}_1) \underline{y} \quad *$$

* FWL thm shows that $\underline{\hat{\beta}}_2$ can be computed from
 $(\underline{I}_n - \underline{P}_1) \underline{y}$ w/ $(\underline{I}_n - \underline{P}_1) \underline{X}_2$

★ from FWL thm, we can deduce the following

$$(I_n - P_1) \underline{y} \sim (I_n - P_1) \underline{x}_2$$

we have

$$(I_n - P_1) \underline{y} = (I_n - P_1) \underline{x}_2 \hat{\beta}_2 + \text{residual vector}$$

$\uparrow$
 $(I_n - P) \underline{y}$

Thus, the residual vector is identical to

$$\underline{y} \sim \underline{x}$$

That is, the orthogonal projection of $\underline{y}$ directly on $\text{span}\{\underline{x}_1\}$ is equivalent to performing iterated projection of $\underline{y}$ on $\text{span}\{\underline{x}_1\}$ and then on $\text{span}\{\underline{x}_2\}$

$$\begin{aligned} (I_n - P_1) \underline{x}_2 \hat{\beta}_2 &= (I_n - P_1) P \underline{y} \\ &= (P - P_1) \underline{y} \end{aligned}$$

★ $\hat{\beta}_2$ describe how $\underline{x}_2$ characterizes $\underline{y}$, after the effect of $\underline{x}_1$ is excluded, and it is different from the OLS estimator from $\underline{y} \sim \underline{x}_2$, unless $\underline{x}_1$ and $\underline{x}_2$ are orthogonal to each other.

